

Title: *The Child Brain Lab – Integrating and innovating meaningful care and research in children with brain-related disorders*

1. General information

1.a Applicants

In case an applicant has more than one affiliation, this is mentioned in the table.

Name*	Institution	Department	Email	Tenured	For ErasmusMC applicants: clinician/ researcher / mix	Role in Flagship
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Planned start date of seed funding: 01/10/2022

Primary Convergence theme: Human Centered Technology and AI for Health

Secondary Convergence theme (if applicable): Improving Health Journeys

Keywords: brain-related disorders, developmental curves, precision medicine, multimodal modelling and AI, biomarkers, co-design, child participation,

1.b Summary

In the Pediatric Brain Center (PBC) professionals of the ErasmusMC-Sophia, TUD and EUR merge their expertise to facilitate optimal development in children with brain disorders. We integrate care and research to generate a platform to rapidly translate new scientific insights and relevant treatment to patients and their families.

As part of the PBC, we are setting up a new facility, the Child Brain Lab (CBL), to assess multiple metrics across domains, ranging from genotype, brain structure and neurophysiology to behavior, quality-of-life, and societal participation in child patients with a variety of brain-related disorders across the life course. We will use a child-centered approach and state-of-the-art techniques (e.g. high resolution EEG, MRI, activity tracking). We aim to include up to 1000 children per year in the first level online assessments (i.e. questionnaires), and 200-400 children per year for the full test battery in the CBL.

Our aims are to:

- 1) Create shared and disease-specific developmental curves with relevant and meaningful outcomes in children with brain-related disorders.
- 2) Develop innovative paradigms and parameters from neurophysiological, brain structure and motor measurements.
- 3) Use advanced data-analyses and AI to identify early (multimodal/multidimensional) biomarkers that predict long term development
- 4) Boost patient engagement by embedding the data acquired in the Child Brain Lab in a life course and precision-medicine framework for use in daily care and to support outcome driven care and shared decision making (SDM) with the so-called Self-Portrait.

Our cross-disorder cohort, our multidisciplinary approach and the facilities in the Child Brain Lab create the unique opportunity to evaluate (novel) interventions which aim at influencing disease mechanisms and improving functional deficits and behavior. Treating disorders as early in life as possible will tap the inherent plasticity of the developing brain, such that interventions will have the most impact.

1.c Public Summary

In het Kinderhersen centrum bundelen professionals van het ErasmusMC, TUD en EUR hun expertise om een optimale ontwikkeling van kinderen met hersenaandoeningen mogelijk te maken. Door zorg en onderzoek te integreren kunnen we nieuwe wetenschappelijke inzichten en relevante behandelingen snel vertalen naar patiënten en hun families.

In het Kinderhersenlab, onderdeel van het Kinderhersen centrum, meten we op een kindvriendelijke manier hoe kinderen op verschillende leeftijden functioneren op belangrijke domeinen; van hersenfunctie en hersengroei tot kwaliteit van leven, gedrag en schools functioneren. In het lab staat het perspectief van het kind centraal. We gebruiken moderne technieken (zoals EEG, MRI, bewegingsmeter) en vragenlijsten om gegevens te verzamelen. Per jaar kunnen tenminste 1000 kinderen meedoen met online vragenlijsten, 200-400 kinderen kunnen het Kinderhersenlab testcircuit doorlopen. Zo bouwen we een uniek cohort waarin overlap tussen verschillende aandoeningen onderzocht kan worden. Doordat we elk kind vanuit veel kanten bekijken, creëren we een prachtige mogelijkheid om behandelingen te testen die gericht zijn op het beïnvloeden van ziektemechanismen, het verbeteren van het functioneren en het beïnvloeden van gedrag. Door aandoeningen zo vroeg mogelijk in het leven te behandelen kan optimaal gebruik gemaakt worden van de plasticiteit van de zich ontwikkelende hersenen, zodat interventies een grotere kans van slagen hebben.

Onze doelstellingen zijn:

- 1) Het maken van normatieve ontwikkelingstrajecten voor betekenisvolle domeinen bij kinderen met hersenaandoeningen om een abnormaal beloop vroegtijdig te detecteren en gepersonaliseerde trajecten te voorspellen.

- 2) Het ontwikkelen van nieuwe uitkomstmaten met neurofysiologische, hersenstructuur en motorische metingen
- 3) Gebruik maken van geavanceerde data-analyses en artificiële intelligentie (AI) om samenhang tussen vroege kenmerken uit verschillende domeinen te identificeren die ontwikkeling op lange termijn voorspellen en effect van (nieuwe) behandelingen evalueren.
- 4) Het stimuleren van patiëntbetrokkenheid door de gegevens uit het Kinderhersenlab in te bedden in de dagelijkse zorg en ter ondersteuning van uitkomstgerichte zorg met het zogenaamde 'Zelf-Portret'.

2.Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

The *Convergence H&T* mission is to tear down the boundaries between different disciplines and institutions with the aim to foster scientific discovery and technological innovation. This Flagship's mission fully aligns with the H&T mission, as professionals from multiple disciplines of the ErasmusMC, TUD, and EUR combine their expertise to form the Pediatric Brain Center (PBC) with the aim to facilitate optimal development in children with brain disorders. As part of the PBC, we are currently setting up the Child Brain Lab (CBL), to longitudinally measure multiple metrics across different levels and domains, ranging from genotype, brain structure and neurophysiology to behavior, quality-of-life, and societal participation in a large group of child patients with variety of brain-related disorders. We will use a child-centered approach and state-of-the-art techniques (e.g. high resolution EEG, MRI, activity tracking) to collect data. We aim to include up to a 1000 children per year in the first level online assessments (i.e. questionnaires), and 200-400 per year for the full test battery in the Child Brain Lab. See Figure-1 for the CBL-battery.

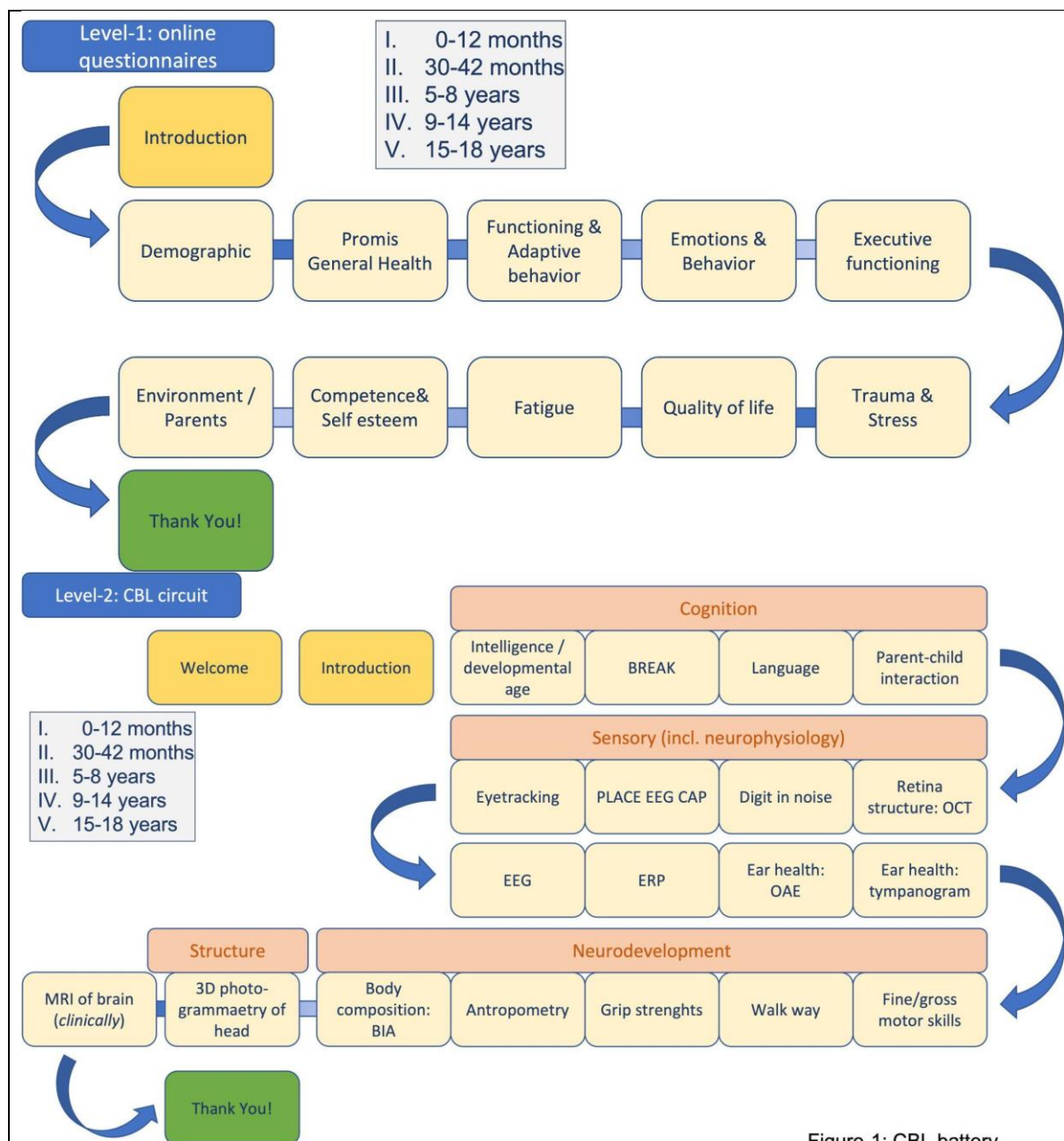


Figure-1: CBL battery

First cross-institution collaborations were initiated in 2019 forming multidisciplinary teams to achieve our goals. The aims of the CBL fit all four H&T aims, and thus has the potential to have a significant and sustainable impact on science, health(care) and society:

Aim-1: Life-long health, from pre-birth to end of life

Through the CBL we will gather high quality long-term follow-up data, from pre-birth (when several brain disorders can already be detected) into early adulthood. The Lab is set up in such a way that results from Level-1 (online questionnaires) and Level-2 (lab visit) are available to the clinician and will be shared with the patient and parents. Sharing relevant findings during the consultation makes it possible to early identify deviations in development (e.g. motor or cognitive development) or problems that prevent recovery (e.g. stress, trauma). Early intervention may be initiated, thereby improving the likelihood of reaching the optimal level of functioning of a child given their disorder-related limits and will assist transition to adult life. Enriching the patient journey with a (lifelong) personal recollection instrument (a 'souvenir' or 'look-back' module of

the self portrait, see Aim-4) will support continued emotional processing of medical experiences while growing up.

Aim-2: Socio-economic equality in health(care)

Data on socio-economic status are collected in Level-1. This will generate information on its impact on the course of disease and level of functioning. Additionally, it will increase insight into targets that may be addressed to either prevent or treat disease. Clarity in communication with child patients through their Self-Portrait will specifically benefit low-SES families.

Aim-3: Prevention and early diagnostics to proactively maintain health

This aim of the Flagship Program lies at the heart of what the CBL aims to achieve, i.e. detect markers, as early in life as possible, that predict outcome or progression of disease and that offer targets for prevention and early intervention to assist the children in reaching their optimal maximum level of functioning. To reach this the CBL will develop new paradigms, analysis methods, and parameters to identify clinical, behavioral, emotional and biological predictors for children with a range of brain disorders from pre-birth to early adulthood.

Aim-4: Individual-tailored precise medical treatment for all

Individual-tailored precise medical treatment will be delivered to patients via two routes. In the short term, children and their parents will undergo additional testing (to what is collected during care as usual) in the CBL, thereby covering all domains reported to be meaningful by families (as identified in a project done by rehabilitation medicine in the PBC). Results from the CBL visit will be communicated with patients and their parents in a so-called Self-Portrait, i.e. a tool for children to map and question their own health situation in personally relevant ways as they grow up, and can be used for clinical consultation and shared decision making and treatment.

In the long term, collecting data on large cohorts and application of machine learning and AI techniques will allow the development of (disorder-specific) normative trajectories, which will not only increase insight in the underlying specific or shared etiology and pathophysiology of disorders, but will also guide diagnostics and prediction of future outcomes for this individual patient.

Our plans cover two of the H&T themes:

Theme-1: Human Centered Technology and AI for Health

The systematic, standardized and longitudinal collection of data in children with different brain-related disorders across a wide range of domains will provide the unique possibility 1) to promote individual health and implement effective prevention and treatment strategies that are shared between children, irrespective of the diagnosis of a child, 2) to detect early signs of disease progression or deviations from normative trajectories through multimodal and multidimensional modeling using AI and machine learning techniques, and 3) to deliver more personalized treatment from the earliest moments and throughout life.

The collected data in Level-1 and 2 of the CBL will provide a unique database which can be used to develop novel technology, including, imaging technology, devices, and artificial intelligence tools, to facilitate new opportunities for prevention, diagnostics, cure and care.

Theme-2: Improving Health Journeys

By starting the CBL assessment in the first year of life, we will contribute to providing an optimal start to life, maintaining optimal development throughout life, predicting relapse or disease, improving treatment outcomes, and optimizing level of functioning and quality of life by the time the children reach adulthood. This will help transform the traditional healthcare system, a 'disease cure system', into a connected system that focuses on the entire health journey throughout life, effectively integrating various formal and informal health care systems with multi-layered data sources.

The CBL aims to engage children and their parents and motivates them to take an active role in decision making, by developing tools to share the findings of the investigations in a child-friendly

and personalized manner. By doing so, we lay the foundations for a more proactive and personalized whole-life approach to health and healthcare. Personalisation will result in improved accuracy of diagnosis, optimized treatments, and minimal side-effects, leading to enhanced resilience, faster resumption of daily life activities, and a longer time-span of healthy and active living. A transition from reactivity to proactivity will empower and motivate the children and their parents to be actively involved in their own health. This is key to improving health and quality of life, reducing the burden on the health care system, and addressing socioeconomic inequalities in health.

2.b Synergy between disciplines

The CBL convergence brings together researchers and specialists from different departments of all three institutes: clinicians and specialists in the field of brain function and brain pathology, imaging and ultrasound, neurophysiology and rehabilitation, researchers in industrial design, biomedical engineering, mathematics and specialists in health management and patient engagement.

Synergy of expertise and experiences between the three contributing institutions is a prerequisite for the CBL to deliver its aims. The CBL will build upon the collaborations that were already in place before the call for flagship proposals was published.

The CBL seeks to involve, in a multidirectional way, the perspectives, the demands and the expertise of all participants from the three institutions in all aims, goals and deliverables. Demand may be driven by patient needs, but also by other drivers such as design, health management or technical inventions. Partners from all institutes will therefore be involved in each work package (see Figure-2). To guide the knowledge exchange, regular brainstorm sessions will be held in large plenary sessions and small focussed working groups.

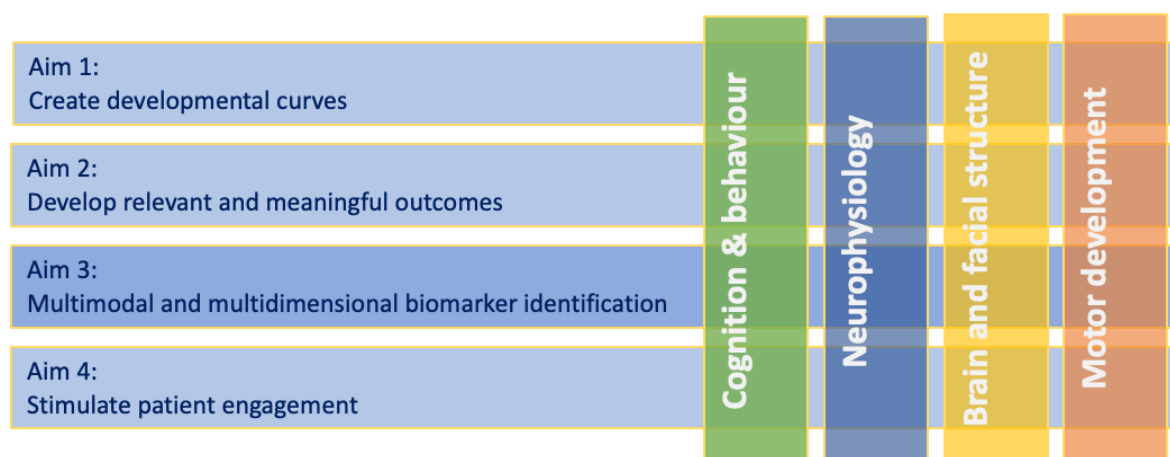


Figure-2: Child Brain Lab's aims

Specifically, three directions for synergy and the urgency to collaborate are defined. First, through the CBL convergence, new devices, paradigms and outcomes will be invented to assess brain physiology and anatomy. This requires translation of test results into relevant parameters, for example through signal analysis of EEG. Second, exploration of the vast amount of data that is produced with complex statistical approaches, such as network analyses, AI or random forest methods, model development and simulations. Third, to improve patient engagement ErasmusMC, TUD-Industrial Design Engineering (IDE) and ESHPM cooperate to deliver a Self-Portrait for children to communicate clinical findings in personally meaningful ways. The Self-Portrait will support the interaction and communication between the triad of child, parents and

clinician. Combining the medical, user-centered and health management perspectives will generate more in-depth insights and a communal medical-research-through-design approach. The CBL in itself creates synergy by its cross-diagnostic and cross-domain/cross-modality approach. We hypothesize that integration of data from different domains and modalities collected over the life course of children with a variety of brain disorders, e.g. relating structural connectivity to functional connectivity, will provide insight in disorder-specific and/or shared pathophysiological mechanisms, predict outcome and help to test efficacy of treatment.

2.c Long term sustainability plan

We set up the CBL for it to stay. To assure its sustainability beyond the 5-year funding period of the Convergence program we dedicate WP6 to sustainability and communication. In WP6, the CBL core team will be responsible for developing a strategy to secure CBL's sustainability through different routes. First, we will continuously identify grant options, explore crowd-funding opportunities and initiate follow-up fundraising with Friends of Sophia, who realized our starting grant. Second, we will start negotiations with health insurance companies based on evidence that the CBL outcomes improve the quality of health care, increase the satisfaction of patients and their families with the provided care, and lead to lower costs in the longer term by identifying problems earlier to increase the window of preventative strategies. Third, the 5-year aim is to build and organize the CBL infrastructure in such a way that it fits the requirements and can apply for the status of core facility of the ErasmusMC/TUD/EUR.

2.d Visibility, attracting talents and new partners

Visibility

The CBL is a unique initiative in the Netherlands and beyond, and as such is a strong brand to create visibility and attention. WP6 will focus on communication, public relations, and dissemination to all relevant stakeholders. As part of our communication strategy, we will set-up an easily accessible website, publish annual reports and offer site-visits during international congresses held in the Netherlands and, on demand, for interested clinicians, researchers and fundraisers. Research output (in lay and scientific literature and congresses) will be presented on behalf of the CBL Research Group.

Student engagement

The CBL convergence provides excellent opportunities for BSc/BA and/or MSc/MA students from the participating institutes to participate in interdisciplinary projects and assist the PhD students, postdocs and researchers. Coordinators and students at EUR, TUD and ErasmusMC will be informed about potential projects. We have received internship applications from students from different disciplines and universities, indicating that they already know how to find us. Students from the ErasmusMC, EUR and TUD will be prioritized where possible. All faculties highly value and have ample opportunities in place for students to engage in CBL projects, e.g. through a bachelor or master thesis, internships and specific design and research courses. The consortium has already benefited from and produced scientific publications (Meulendijks 2021) of such student work. We will recruit from: ESHPM, TUD/IDE, TUD/Biomedical Engineering (BME), Technical Medicine (LDE), ErasmusMC/NIHES, ErasmusMC/Medicine and Psychology/Pedagogic. Educations admit between 100-300 students per year. Study programme directors within the CBL consortium are Schouten (Biomedical engineering) and Prof. Harlaar (Technical medicine).

We will safeguard the capacity to supervise large numbers of students by producing a 'project calendar' for students to subscribe to, including a coaching plan (e.g. a series of projects linked to

each postdoc or PhD-trajectory), a student peer-support group and invitations for linked activities, e.g. acting as clients in courses using our themes under their own primary mentorship.

Talent and partners

In an attempt to find the best PhD students and postdocs, we will recruit via national (e.g. Academic Transfer, LinkedIn) and international platforms (mailing lists/websites of relevant scientific societies/conferences) in addition to the websites of our institutions.

The CBL will be an interesting partner for institutes and companies that develop new diagnostic tools and new treatments, as the lab is equipped to provide excellent validation and test options. Also for partners that develop tools to enhance patient engagement, such as interactive apps, cooperation with the CBL will be attractive. These potential partners will be actively approached by the CBL to build consortia that can apply for large national and international funding opportunities.

The CBL will organize Round Tables with representatives of other Flagships, specifically, *Deep Medical Imaging* (to exchange expertise in brain imaging biomarkers) and *iMED* (to link with their fourth aim to implement precision medicine in the clinic).

External funding

Researchers from all participating departments will apply for funding to address research questions that fall within the mission of the CBL. We have identified the following opportunities:

- Zorginstituut NL: Samen beslissen met kinderen
- NWO-ORC (project idea submitted in 2022)
- CEN 3d measurements in 1800 normal children (european grant, last round)
- SSWO
- NWO-XS
- NWO Open competition Domain Science
- NWO Open competition SSH (currently no open call)
- VENI-VIDI-VICI scheme
- Hersenstichting Hoofdzaken
- Hersenstichting Praktijkgericht onderzoek

Additional funding opportunities include patient wellbeing foundations, insurance companies, and partly in-kind contributions of partners in development of the Self-Portrait.

2.e Long term societal impact

Better understanding of the consequences of brain-related disorders will help children to optimize how they function in the real world by earlier intervention aimed at e.g. improving school performance or socio-emotional development. Optimizing children's societal participation and quality of life will reduce health care costs in the long term.

Increasing engagement of patients in shared decision making will improve the quality and cost-effectiveness of medical care. Through our research on the Self-Portrait, we will uncover strategies to involve a variety of children in more intense engagement with their care path, which will provide a blueprint for other medical domains to follow.

The rapid expansion of DNA-testing in the last 5-10 years has greatly increased the number of genetic causes found in children with developmental problems and intellectual disability. While this has enabled counseling of families, there is a distinct lack of information on prognosis and treatment options. Furthermore, from fundamental research, promising options of genetic (e.g. <https://www.rnatherapy.nl/>) and small molecule treatment are ready for translation to the clinic.

We need natural history information and well selected outcome measures to design studies to test efficacy. The CBL is in a unique position to provide such information and measures and thus accelerate this transfer of health research into practice. The natural history cohorts and the normative trajectories will be valuable and shared and expanded nationally and internationally where possible.

3. Flagship proposal

3.a Ambitions

The CBL mission is to enable patients to fully employ their potential to participate and function in society at the highest level possible. Our ambition fully aligns with the Convergence H&T research because we need the expertise of the medical and health sciences, life sciences, technical and data sciences, social sciences and humanities to make our healthcare system for children with brain-related disorders more proactive, precise, participatory, and labor friendly.

Our main innovation lies in realizing a life course perspective on mental, physical and social health of children with brain-related disorders. The heterogeneous and multimodel/multi-dimensional datasets, with time as an additional dimension, present a challenge that requires innovative analytical methods. Integrating mental, physical and social levels of information will generate insight in patterns that predict future functioning, which cannot be identified by looking at each individual parameter. Co-design and co-research are core features of the CBL to shape patient participation and engagement with the aim to empower patients and parents.

3.b Aims and feasibility

Aims:

- 1) Create shared and disease-specific developmental curves with relevant and meaningful outcomes across four domains (cognition & behavior, neurophysiology, brain & craniofacial structure, and motor development) in children with brain-related disorders, both from already established and validated parameters and new-to-develop parameters.
- 2) Develop innovative paradigms and parameters from neurophysiological, brain and craniofacial structure and motor measurements.
- 3) Identify complementary information or trends in the multimodal/high-dimensional data that inform pathophysiology and predict long term functional development by applying machine learning algorithms and deep learning networks.
- 4) Boost patient engagement by embedding data across data domains in a life course and precision-medicine framework for use in daily care and to support outcome driven care and shared decision making (SDM) with the so-called Self-Portrait.

Our cross-disorder cohort, multidisciplinary approach and the facilities in the CBL create the unique opportunity to evaluate (novel) interventions, which aim at influencing disease mechanisms and improving functional deficits and behavior. Treating disorders as early in life as possible will tap the inherent plasticity of the developing brain, such that interventions will have the most impact.

Feasibility - staff and budget:

Longitudinal data collection will take place at two levels (Figure-1). Level-1 consists of an online assessment of (mostly validated) self-reported and parent-reported questionnaires on mobility, daily activities, social-cognitive functioning, and health-related quality of life. Data collection at Level-2 takes place in the three rooms of the CBL, respectively dedicated to 1) assessments of cognitive functioning, language, and parent-child interaction, 2) neurophysiology and craniofacial structure, among others, EEG, eyetracking, 3-D photogrammetry, and OCT and 3) balance and gait analysis and activity tracking. In addition, a harmonized collection of optimized T1 and T2-weighted magnetic resonance imaging (MRI) scans is currently being set up to become part of care as usual.

We will start Level-1 data collection in June 2022 and will recruit 1000 children per year. We expect the first patient to visit the test facility for a Level-2 assessment in September. We have identified the disorders we want to start with and discussed with clinicians the feasibility of these numbers.

Staff	Institute	Duration	Background	WP(see Q3c)
New staff				
postdoc	Erasmus MC	5	psychology	1
postdoc	Erasmus MC/TUD/Rijndam	5	biomechanical-engineering, rehabilitation	1&2-motor
postdoc	TUD/ErasmusMC	5	biomechanical-engineering	2-neurophysiology

postdoc	TUD/ErasmusMC	4	biomechanical-engineering	2-structure
postdoc	TUD	4	mathematical/statistical	3
PhD	TUD/EUR	4	design research for children's wellbeing	4
app-developer	external			4
communication advisor	external			5
Cash matching (yrs 1-2, part of 'use of CBL facility' in budget)				
coördinator	Erasmus MC			CBL
datamanager	Erasmus MC			CBL
test assistant	Erasmus MC			CBL
In kind matching				
medical specialist	Rijndam	5(0.1 fte)	Rehabilitation	1,2
clinical linguist	Rijndam	5(0.1 fte)	Rehabilitation	1
postdoc	Erasmus MC	7m(0.5 fte)	Rehabilitation	1,2
Research Suite and Data&Analytics				
dashboard-developer	Erasmus MC			5

Strategy to involve students: see section 2d.

Risks & Mitigations:

Patients and their parents are not motivated to participate in the CBL.

- Report back clinically relevant findings during the first consultation after the collection of Level-1 or Level-2 information.
- Communication tools to provide patients and their parents with information about our mission and aims (website, leaflets, explanations during consultations).

-Seek active engagement of patients and their parents through interactive sessions to collect their views related to, e.g., outcome-driven health care and design solutions.

Data collection is too slow.

-Students will assist in data collection and data quality checking (section 2d).

-Commitment of dedicated multidisciplinary teams to support the CBL is large.

Clinicians and parents may be afraid to share sensitive health information with children.

-Start with a subset of relatively straightforward outcomes as demonstrators to build trust with stakeholders about sharing and filtering strategies for each outcome type.

-Develop the most optimal strategy of sharing health information with children through co-design.

The wide variety in developmental age of patients may hamper participation in co-design.

Although this is part of the research challenge, we have identified several strategies (applying alternate forms of communication, including proxies such as former child patients, triangulating with parents) to mitigate these risks. The consortium has successfully piloted the convergence approach on this matter (Figure-3).

Funding

See section 2d.



Figure-3: examples of co-design materials, preliminary designs and research outcomes of the Self-Portrait.

3.c Work plan

Aim	Deliverables

WP1: Create developmental curves with existing and validated instruments and new to develop parameters	
Estimation and analysis of normative developmental trajectories of relevant parameters. This requires integration of parameters collected in time. First exploration can be done on existing datasets from subpopulations of CBL from Level-1 and Level-2: - Cognition, behavior and language - Mobility and daily activity	Normative developmental trajectories for various parameters that are shared between or specific for different brain-related disorders.
WP2: Identify new paradigms and parameters with neurophysiology, brain structure and motor analysis.	
a. Neurophysiology (Level-2) <u>EEG</u>: Analyses of functional connectivity to obtain correlations between EEG patterns and outcome parameters. <u>ERP</u>: Prediction of variation in neurodevelopmental trajectories from auditory event-related potentials (ERPs; in collaboration with WP1). Analyses of EEG patterns, event related potentials and responses to auditory stimuli for diagnostic and prognostic purposes. Analyses of how hearing acuity impacts on communication development and disorders. Analyses of sensorimotor trajectories that are based on eye- and hand-movement tasks combined with cognitive behavior to quantify the development of functional brain networks.	Pipeline for pre-processing and analysis of neurophysiological data, including quantified EEG and ERP parameters. Workflow for training machine learning algorithms on both neurophysiological data. Documentation of cross-cohort differences in EEG and ERP recordings. Neurophysiological network biomarkers to identify children at risk at an age when intervention is most likely to benefit them.

b.Brain and craniofacial structure (Level-2 & Clinically acquired MRI-scans) <u>Brain Structure (MRI)</u>: Using AI and machine learning to quantify brain developmental trajectories in relation to functioning in different domains and linking anatomy to functional network connectivity. <u>Facial shape (3-D photogrammetry)</u>: Skull shape and face analysis for diagnostic purposes, genotype phenotype coupling, growth analysis and posttreatment evaluation. Collect values from healthy populations. <u>Retinal characteristics (OCT)</u>: Assessment of retinal nerve cell layer and correlation to MRI properties and clinical/functional outcome.	Normative brain/facial/skull developmental trajectories that are disorder-specific or shared between disorders, that allow early identification of deviation and prediction of outcome. Processing and analysis pipelines for automatic extraction of anthropometric and shape parameters as well as personalized modeling. Retinal biomarkers for brain structure and neurophysiological outcomes
c.Motor functioning (Level-2) Validate new assessment techniques for <u>balance</u> and <u>gait</u>, aiming to detect early indicators of problems in motor development. These techniques focus on developmental milestones, e.g. sitting, standing, walking (Bekius 2021, Blok 2022).	Precise and easily applicable methods of trunk control and gait ability for early diagnostics of motor disability, using balance profiles and temporal-spatial Gait indexes. 12 months: candidate tests selected from trials in a typically-developing group. 24 months: reliability, validity of the tests known for the patient group.
WP3: Multimodal/multidimensional biomarker identification	
Apply machine learning algorithms and deep learning networks to identify complementary information or trends in the multimodal/ high dimensional data, using machine learning (random forest method) and convolutional neural networks. Overfitting will be prevented by feature selected algorithms.	Workflow for training machine learning algorithms on multimodal data. Early biomarkers for functional outcome ("will my child be able to walk?") based on correlations between e.g. MRI and EEG findings.

WP4: Patient engagement	
Using the LIFE method (Senges, 2021): Identify the age-dependent needs of the triad of child, parents and health care provider with respect to the four data themes. Define which metrics from the four data themes are suitable for sharing with children and co-design prototypes of the Self-Portrait in the <i>Life phase</i>. Investigate through co-creation which design principles support patient information and shared decision making. Perform age-specific product development and test the Self-Portrait through iterative design processes in the <i>Innovate</i> and <i>Field test phases</i>. Evaluate the effect of the Self-Portrait on satisfaction of the care path in the <i>Evaluate phase</i>.	A Self-Portrait for patients visiting the CBL, that empowers them to engage in shared decision making and to guide communication with their parents and clinician. Evidence-based design features that support patient engagement or assist in communication in the triad of child, parents and clinician. Description of the relation between design features of the Self-Portrait and patient information and shared decision-making to be used for later effects research. Documented advances in methodology of co-design with children in healthcare.
WP5: Data Handling and Infrastructure	
Development of a user-friendly infrastructure to foster smooth inclusion of new participants, high-quality data collections, safe data storage and efficient data delivery to the different end-users (clinicians, patients and parents, researchers). This aim converges with the roadmap of the Research Suite program of the ErasmusMC. This will accelerate the development and enable tailored features for the CBL. Figure-4 provides a visual representation of the intended data infrastructure of the CBL.	A data infrastructure that complies with the local, national and international legislations, GDPR and FAIR principles. An automated data infrastructure but with human checks on crucial points of the process as a quality assurance. Informed Consent manager to assure the rights of the participants are fulfilled. Cohort manager, to assist in the process of recruitment, appointment making and status tracking. Clinical dashboard to support clinical decision making and value-based healthcare. Self-Portrait to engage patients with the CBL.

	Castor database with trial-ready research data. Implementation of the Digital Research Environment (anDREa) to facilitate data sharing between EUR, TUD and ErasmusMC (and beyond).
WP 6: Communication and Sustainability	
Defining ways to secure funding by identifying grant options, starting negotiations with health insurance companies, exploring crowd funding possibilities and setting up followup activities with Friends of Sophia. A communication plan will be defined and implemented.	Sustainability plan Communication plan ● Website ● Annual report ● Site-Visits ● Social media exposure ● Matchmaking events (NWA or Horizon)

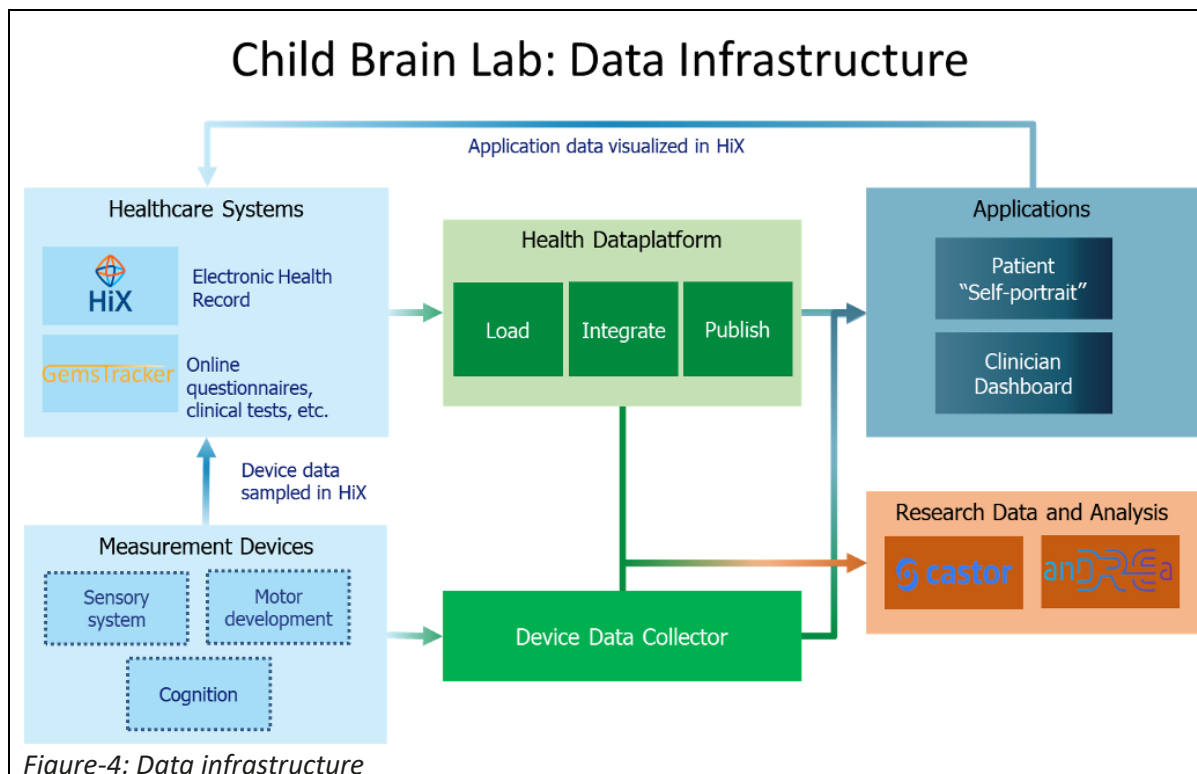


Figure-4: Data infrastructure

3.d Alignment of the program with relevant national and international research agendas

ZonMW and Hersenstichting launched "programma Hoofdzaken" for 2023. This program defines three major goals for knowledge and data in the fields of psychiatry, psychology, neurology and neuroscience:

- involvement of patients and their families as partners in all phases of research
- cross-disease research to increase insight in mechanisms, symptoms and solutions,
- collaboration on all levels and disciplines to share knowledge, techniques and data.

The CBL consortium registered an initiative for Dutch Research Agenda (NWA-ORC 2022): our project aligns with four research routes on the agenda:

- Health care research, sickness prevention and treatment
- Child and adolescent development, upbringing and education
- NeuroLabNL: the ultimate living lab for brain, cognition and behavioral research
- Personalized medicine.

Partners as described in section 2.d will be approached.

Horizon Europe, Cluster 1-Health, will publish continuous calls for health research. In June 2022 the opening is planned for the call: Implementing digital services to empower neuroscience research for health and brain inspired technology via EBRAINS.

For 2022, Zorginstituut Nederland chose the theme: "Deciding together with children". Subsidy is available for projects about learning from outcome measures in shared decision making with children.

4. Consortium

4.a Track record of the consortium members

The CBL-consortium consists of junior and senior researchers from the 3 institutes, 16 departments and Rijndam Rehabilitation. In the past 5 years the participants of the CBL consortium supervised >400 master students and >95 PhD students. Together they raised >22ME of funding, including multiple grants from NWO, ZonMw, NIH and Horizon Europe. A grant from CEN-cenelec has been committed.

ErasmusMC

Pediatric Brain Center (PBC)

Clinicians and researchers of the PBC have been cooperating since 2018 to integrate care and research for children with brain-related disorders. Care is organized in multidisciplinary care-paths for a wide range of congenital and acquired disorders, such as autism, cerebral palsy, developmental disorders. The center focuses on (technical) innovations and fast implementation in care. Important projects are new medications (e.g. small molecules, RNA therapy), deep brain stimulation for epilepsy, intrauterine closure of myelomeningocele, serious gaming for emotional disorders in adolescents. The CBL provides the necessary set-up to evaluate these innovations and realize further improvements.

Van Veelen: Lead of the PBC. Neurosurgeon, focus on implementation of new techniques and long-term evaluation. Board member European Society of Pediatric Neurosurgery, workstream lead craniosynostosis ERN Cranio.

Prof Van Haren (brain development and psychopathology): She merges expertise from psychiatry, neuroimaging, genetics, and cognition to better understand the etiology underlying (risk of) severe mental illness across the lifespan. Van Haren received VIDI and NIH funding and was recently awarded a Horizon Europe grant. She is setting up a CBL neuroimaging program with departments who treat critically ill neonates undergoing surgery (i.e. SHARING/CRUCIAL/SMILE).

De Wit: neurologist and specialist in genetic neurocognitive syndromes and epilepsy. Clinical director of ENCORE expertise center for neurocognitive diseases and PI in observational studies and phase 2/3 medication trials (Overwater 2019, Bindels 2020). Research within ENCORE addresses Natural History studies of the Rare Neurodevelopmental Disorders, e.g. GRIN-GRIA and CAMK2 (Veenma) and neuropsychological and psychiatric aspects of rare brain diseases (Rietman). For various rare diseases findings were translated into a clinical application: "Talking with Children about....".

Roebroek: associate professor rehabilitation medicine with expertise on childhood neurodisability, long term development in several life areas and life span approach (vanGorp 2018, Schmidt 2020). Her research gave a boost to transition programs and young adult teams in rehabilitation. Past chair of Lifespan Care Committee of the American Academy of Cerebral Palsy and Developmental Medicine.

Current initiatives: Collaborative projects from neurology and TU Delft are "EEG-based machine learning algorithm for the neurological prognosis of children after OHCA", "Machine learning systems for combining multimodal data on the intensive care unit" (vd Berg) and "Evaluation of

ERP measurements as a marker for later linguistic and cognitive development” (vd Schroeffer). Other clinical and research expertise focuses on participation and activity, nutrition and metabolism, hearing and speech evaluation, motor and gait-analysis (Toussaint) and the development and implementation of value-based health care. Participants are active in national and international consortia, such as National consortium development screening instrument youth health care (Joosten), DOOFNL, SEED, PACE (vd Schroeffer), NIH CARE project for craniofacial Microsomia (Koudstaal) and European Rare diseases Networks (e.g. ERN-Cranio (Prof Mathijssen main lead), ERN-Arnica).

TU Delft

Industrial Design Engineering

The faculty of Industrial Design Engineering has a long history of human-centered design research for healthcare and for specific user groups (e.g. hospitalized children, children with impairments). Recent and current projects from consortium members include:

- Design for children’s motivations, play and wellbeing (Play Well Lab)
- Child-centered design in medical and hospital context (e.g. communication tools for hospitalized children (Meulendijks 2022); EEG procedures; Gait analysis; in-hospital play for health)
- Stakeholder involvement in hospital context (nurse participation in work environment improvements)

Gielen: Expertise in research on children’s perspectives, co-research and co-design with children (Gielen 2013, 2020; van Doorn 2016), resulting in several tools and products addressing children’s health and co-design (Your Turn (Gielen 2020), Lumies cubes (Lomas 2021)).

Prof Stappers (design techniques): Expertise in methods of Co-design/context-mapping and Research through Design (Stappers 2012, 2017) applied in medical and health settings such as “Meedoen=Groeien” for Princess Maxima Center for Pediatric Oncology, stimulating physical activity in young cancer patient (Boon 2020), and “Familiegesprekken”, stimulating awareness of and planning for old age and other care.

Huysmans: Computer science and digital human modeling; <https://dined.nl/en>. Expertise in ergonomics and human-centered design. Translation into clinical practice, e.g. redefinition of headforms for cycling helmet testing. A CEN-cenelec grant has been committed.

Biomechanical engineering

Schouten: Neuromechanics and motor control. Electrophysiological signal analysis (EMG/EEG). (co)PI in multiple projects on mobility (e.g. MoveOn, a novel balance-testing device to improve mobility after stroke), and Neurocontrol: neuromodulation by continuous and integrated monitoring and treatment.

vd Ruit: Neuromechanics: software application for TMS mapping. Multidisciplinary projects ReArm (TU, Erasmus, Rijndam), Brain@home (TU, LUMC, Universiteit Leiden).

Poggensee: Expertise on human movement science/biomechanics, assistive devices, and robotics. She has experience quantifying and classifying human movement using statistical and machine learning methods.

Prof Vallery (mechanical engineering): Expertise on bipedal locomotion, balance and human-robot interaction, focus on developing technology that empowers individuals with locomotor disabilities (Paiman 2016, Plooi 2021). For example: fall detection algorithms for wearable airbags for fall prevention. Keynote Speaker on tech-transfer at the European Robotics Forum.

Prof Harlaar (clinical biomechanics): Focus on the clinical application of new technologies, e.g. his multimedia movement analysis system became the national standard (Flux 2020), and co-developer of the GRAIL, a clinical VR gait system.

Mathematics

Prof Jongbloed: Large expertise in statistics with many multidisciplinary projects resulting in clinical applications, e.g. Nederlands Forensic Institute (quantification of strength of evidence), KNMI (extreme event prediction), LUMC (data integration).

ESHPM (EUR)

The department of Health Services Management & Organization focuses on management and organization to provide the best value for patients/clients and professionals in healthcare. Research themes include value-based healthcare, co-creation, and patient engagement (projects: shared decision-making in daily practice; BUZZ2 on an outcome-based payment model in maternity care; ACCRA, to advance Robotics-based solutions for extending active and healthy aging).

Tableau: assistant professor in health services innovation and co-creation. Research interests are healthcare innovation, product and service development and multi stakeholder co-creation (Gyrard 2021, Fiorini 2019, Tableau 2016). She is an expert in the LIFE-methodology for co-design in value-based care (Senges 2020).

Porte: assistant professor in technological innovations in healthcare, with expertise in clinical technology and informatics. Focus on the use of data in healthcare and evaluation of home-monitoring tools. She teaches in courses where data and technology solutions are co-designed for healthcare.

Prof Ahaus: His research interests are value-based health care (Steinmann 2021, Ahaus 2018). He is part of the NFU expert panel on this topic since 2015, and developed and delivered an Advanced Training Program Value-based Health Care with all UMCs.

Prof Stiggelbout (ESHPM&LUMC): Focuses on individual doctor-patient decision making, involving doctor-patient communication. This includes risk communication, as a constituent of shared decision making, and normative and ethical aspects of medical decision making. Her research is methods- and implementation-focused.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Each WP consists of a combination of participants with complementary expertise.

participant	role/expertise
WP1: create developmental curves with existing and validated instruments and new-to-develop parameters	
Dieleman/Dulfer/Rietman	Psychology, longitudinal testing in children with developmental disability
Heydenrijk/Roebroeck/Pangalila/vd Meulen/Dekkers/Duyster	Meaningful outcomes, life span approach Language/speech and motor rehabilitation and evaluation
De Graaff/Veenma/Joosten/Olieman	Internal medicine and pediatrics for rare (developmental) diseases activity and metabolism
De Wit	Expertise developmental disorders
WP2: Identify new paradigms and parameters with neurophysiology, brain structure and motor analysis	
a. Neurophysiology	
de Wit	Neurology, epilepsy
vd Berg/Straver	Neurophysiology, expertise in signal analysis EEG
Schouten/vd Ruit	Biomechanical engineering, expertise in neuromechanics, electrophysiological signal analysis EEG. Machine learning application.
Geleijnse/Vroegop/Schroeff	Technical medicine, expertise in hearing, brainstem evoked potential, evoked-related potential (ERP)
Pel/Dieleman	Psychology, expertise in eye-tracking
b. Brain and craniofacial structure and anatomy	
Van Haren	Brain development and psychopathology, correlation to cognitive/behavioral development and genetics
White	Neuroimaging Generation-R

Roschupkin/Huysmans	Computer modeling, MRI, 3d-photogrammetry and genetics Human centered design, anthropometry, statistical shape modeling
van Veelen/Mathijssen/Koudstaal	Neurosurgery, craniofacial surgery, defining relevant outcome parameters related to raised Intracranial pressure and skull growth
Loudon	Ophthalmology, OCT
c. Motor functioning	
Roebroeck/Duyster/Pangalila	Rehabilitation, Motor development, gait analysis
Poggensee/Vallery/Horemans/Harlaar	Gait analysis, biomechanical engineering
Kempink	Orthopedic surgery, cerebral palsy
WP3 Multimodal/multidimensional pattern recognition and biomarker identification	
Van den Berg/Schouten	Multimodal analysis
Jongbloed	Mathematics and statistics: Advanced analysis: explainable AI, machine learning, convolutional neural networks
WP4 Patient engagement	
Gielen/Stappers/Porte/Tabeau/Ahaus/Stiggelbout	Human-centered design research Co-research and co-design with children Patient engagement in care, SDM
Van Veelen/de Wit/ Roebroeck/van Haren	Clinical user-perspective, meaningful outcomes, parameter development
WP5 Data-handling and Infrastructure	
van Haren/Roschupkin/Schouten	Extensive experience in handling large databases, provide user-perspective
Schrijvers	Data-architect, research suite program (erasmusMC facility responsible for data extraction from patient dossiers)
WP6 Communication and Sustainability	

van Veelen	lead PBC
van Haren/de Wit/ Roebroek	Core team CBL, experience in fund-raising
Gielen/Tabeau	leads flagship

4.c Involvement of partners

Rijndam Rehabilitation

Rijndam provides the necessary insight in concepts of functioning in society. This group has experience in building developmental curves for meaningful outcomes and national longitudinal follow-ups in cerebral palsy and childhood aphasia. Rijndam conducted a project to identify meaningful outcomes for patients in the PBC, which was the basis for the CBL's battery (Heydenrijk 2021). Rijndam contributes to the set-up and further development of the motor assessment, trunk support and gait-analysis, and with expertise on speech/language development.

Generation-R

The CBL-battery is harmonized, where possible, with Generation-R to enable comparison to healthy controls.

Partners of TU Delft

Support is available from TU-wide institutions: Sports Engineering Institute (health technology), Wetenschapsknooppunt and Bètasteunpunt (co-design with children in Primary and Secondary Education). These institutions provide advice, access to their network and external funding applications.

Patient representative groups

The CBL-participants connect to patient representative groups to share information, discuss research agendas and collaborate in research projects. Examples of collaborations with patient representative groups are NFVN (neurofibromatose vereniging Nederland), STSN (Stichting tubereuze sclerose Nederland), vASN (vereniging Angelman syndroom Nederland), Stichting Amy and friends, MS vereniging Nederland, EMB Nederland, CP Nederland, Stichting Hersenletsel.

4.d Governance, organization and collaboration strategy of the consortium

Governance

The daily organization of the CBL is in the hands of a Steering Committee (SC) consisting of the three leads of the CBL flagship (van Veelen/Gielen/Tabeau) and the CBL core-team (van Veelen/van Haren/de Wit/Roebroek). The SC is responsible for the set-up of the lab, the financial control, METC and data-handling. The SC oversees the whole of research projects and funding applications, prevents overlap and ensures alignment with the aim and vision of the CBL and decides on in and -external requests for data (together with the relevant PIs). They meet every 2 weeks.

All participants together form the Working Group (WG) which consists of clinicians, researchers, technical specialists, and support staff. They are responsible for involving patients, designing standard operating procedures, producing research questions and research output, and supervising PhDs, postdocs and master/bachelor students. They work together in focused work packages. Plenary meetings with all participants will be organized every 2 months to exchange progress that was made in the smaller work packages.

An academic board is formed consisting of the heads of all participating clinical departments. They provide solicited and unsolicited advice, are annually informed about the progress, and approve the annual plan.

Collaboration with other flagships

Synergies exist between our flagship and several other flagships that will be submitted in the current call. First, we link with our collaborators in the Deep imaging flagship to explore the potential of using our brain imaging data in their new technologies. Second, we align with the iMED flagship to realize their aim-4 (implementing precision medicine in the clinic).

Annexes

I Budget table (including the external funding table)

II Letters of intent and letters of commitment

- a. *Letter of commitment from the Vrienden van Sophia*
- b. *Letter of commitment from Rijndam Rehabilitation*
- c. *Letter of commitment from Department of Rehabilitation*

III Letter of support from the department head of the main lead.

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Budget formulier: Convergentie Flagship Call

Flagship:	Child Brain Lab
Flagship acronym:	Child Brain Lab
Naam Penvoerder:	M.L.C. van Veenen
Startdatum en looptijd van het project:	1-10-2022 => looptijd 5 jaar
Totaal project budget:	€ 4.327.160,00

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind		
Onderzoeksorganisaties	postdoc	1,0	01-10-2022	30-09-2027	Postdoc	€ 4.700	€ 423.000	€ 18.750,00	€ 77.000,00	€ 80.500,00	€ 85.750,00	€ 90.500,00	€ 70.500,00	€ 423.000	€ 0	€ 423.000,00	€ 0	€ 423.000,00				
	postdoc	1,0	01-10-2022	30-09-2027	Postdoc	€ 4.700	€ 423.000	€ 18.750,00	€ 77.000,00	€ 80.500,00	€ 85.750,00	€ 90.500,00	€ 70.500,00	€ 423.000	€ 0	€ 423.000,00	€ 0	€ 423.000,00				
	postdoc	1,0	01-10-2023	30-09-2027	Postdoc 2	€ 4.660	€ 335.500	€ 19.750,00	€ 78.000,00	€ 82.500,00	€ 87.000,00	€ 88.250,00	€ 335.500	€ 0	€ 335.500,00	€ 0	€ 335.500,00					
	PhD student	1,0	01-10-2022	30-09-2026	Phd Student	€ 3.521	€ 253.500	€ 14.375,00	€ 58.500,00	€ 62.500,00	€ 66.375,00	€ 51.750,00	€ 51.750,00	€ 253.500	€ 0	€ 253.500,00	€ 0	€ 253.500,00				
	postdoc	1,00	01-10-2023	30-09-2027	Postdoc 2	€ 4.660	€ 335.500	€ 19.750,00	€ 78.000,00	€ 82.500,00	€ 87.000,00	€ 87.000,00	€ 88.250,00	€ 335.500	€ 0	€ 335.500,00	€ 0	€ 335.500,00				
						€ -	€ 0								€ 0	€ 0	€ 0,00	€ 0	€ 0,00			
						€ -	€ 0								€ 0	€ 0	€ 0,00	€ 0	€ 0,00			
						€ -	€ 0								€ 0	€ 0	€ 0,00	€ 0	€ 0,00			
						€ -	€ 0								€ 0	€ 0	€ 0,00	€ 0	€ 0,00			
						€ -	€ 0								€ 0	€ 0	€ 0,00	€ 0	€ 0,00			
Ondemeningen	Ondemening 1	App ontwikkelaar	1,00	01-10-2022	30-09-2025	€ -	€ 100.000	€ 8.333,33	€ 33.333	€ 33.333,33	€ 25.000			€ 100.000	€ 0	€ 100.000,00	€ 0	€ 100.000,00				
	Ondemening 2					€ -	€ 0							€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Ondemening 3					€ -	€ 0							€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Ondemening 4					€ -	€ 0							€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Overige (private) partners	Rijnham	Diverse personeelsleden				€ -	€ 306.000	€ 7.935,00	€ 15.300,00	€ 13.225	€ 1200	€ 1200	€ 1200	€ 4900	€ 0	€ 306.000	€ 0	€ 306.000,00				
	REVA Erasmus MC	Postdoc funding				€ -	€ 21.160							€ 21.160	€ 0	€ 21.160,00	€ 0	€ 21.160,00				
	Naam 3					€ -	€ 0							€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Naam 4					€ -	€ 0							€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Totaal								€ 2.620.660	€ 86.893	€ 15.300	€ 375.658	€ 61.200	€ 493.333	€ 61.200	€ 613.625	€ 61.200	€ 497.250	€ 61.200	€ 348.000	€ 45.900	Totaal loonkosten	€ 2.620.660

2.	Kosten van materialen en hulpmiddelen:							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Materiaal / hulpmiddel	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties	Translatiegegevens personeel	€ 19.000												€ 19.000	€ 0	€ 19.000,00	€ 0	€ 19.000,00				
	Research Suite	€ 25.000	€ -	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 25.000	€ 0	€ 25.000,00	€ 0	€ 25.000,00				
	Website / social media	€ 20.000		€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 5.000	€ 20.000	€ 0	€ 20.000,00	€ 0	€ 20.000,00				
	Co-design sessie	€ 40.000		€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 8.000	€ 40.000	€ 0	€ 40.000,00	€ 0	€ 40.000,00				
	Licenties	€ 300.000		€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 60.000	€ 300.000	€ 0	€ 300.000,00	€ 0	€ 300.000,00				
														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Training Postdocs	€ 50.000	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 12.500	€ 50.000	€ 0	€ 50.000,00	€ 0	€ 50.000,00				
	Training PhD	€ 5.000	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 2.500	€ 5.000	€ 0	€ 5.000,00	€ 0	€ 5.000,00				
	Monitoring	€ 15.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 15.000	€ 0	€ 15.000,00	€ 0	€ 15.000,00				
	Datamanagement	€ 10.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 10.000	€ 0	€ 10.000,00	€ 0	€ 10.000,00				
Communication material	€ 10.000	€ 4.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 3.000	€ 10.000	€ 0	€ 10.000,00	€ 0	€ 10.000,00					
Round tables with other flagships	€ 5.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 2.000	€ 5.000	€ 0	€ 5.000,00	€ 0	€ 5.000,00					
													€ 0	€ 0	€ 0,00	€ 0	€ 0,00					
Dashboard developer => Data analytics	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 25.000	€ 0	€ 25.000,00	€ 0	€ 25.000,00				
													€ 0	€ 0	€ 0,00	€ 0	€ 0,00					
													€ 0	€ 0	€ 0,00	€ 0	€ 0,00					
													€ 0	€ 0	€ 0,00	€ 0	€ 0,00					
													€ 0	€ 0	€ 0,00	€ 0	€ 0,00					
Organisatie 4	Use of child brain lab facility and infrastructure	€ 1.077.500	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 250.000	€ 77.500	€ 1.077.500	€ 0	€ 1.077.500,00	€ 0	€ 1.077.500,00				
Ondemeningen														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Ondemening 1														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Ondemening 2														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Ondemening 3														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Ondemening 4														€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Overige (private) partners	Naam 1													€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Naam 2													€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Naam 3													€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
	Naam 4													€ 0	€ 0	€ 0,00	€ 0	€ 0,00				
Totaal								€ 1.691.500	€ 0	€ 0	€ 379.000	€ 0	€ 383.000	€ 0	€ 349.500	€ 0	€ 346.500	€ 0	€ 174.500	€ 0	Totaal kosten materialen en hulpmiddelen	€ 1.691.500

3.	Kosten van gebruik van machines en apparatuur							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Machine / apparatuur			Prijs per gebruikseenheid	Gebruiks eenheden	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties	Organisatie 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 3																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 4																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
Ondemeningen	Ondemening 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 3																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 4																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
Overige (private) partners	Naam 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Naam 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Naam 3																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Naam 4																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
Totaal								€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	Totaal kosten machines en apparatuur	€ 0

4.	Aan derden verschuldigde kosten							2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
	Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind	in cash	in kind			
Onderzoeksorganisaties	Organisatie 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 3																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Organisatie 4																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
Ondemeningen	Ondemening 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 3																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Ondemening 4																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
Overige (private) partners	Naam 1																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Naam 2																€ 0	€ 0	€ 0,00	€ 0	€ 0,00	
	Naam 3																					

5.

Publicatie, reis- en verblijfkosten			2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner
Partner die de kosten maakt	Naam derde, omschrijving kosten	Kosten	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	
Onderzoekorganisaties																	
Organisatie 1	Risikosten postdocs	€ 50.000			€ 10.000		€ 10.000,00		€ 10.000		€ 10.000,00		€ 10.000,00		€ 50.000	€ 0	€ 50.000,00
Organisatie 2	Risikosten PhD	€ 5.000			€ 2.000		€ 2.000,00		€ 1.000						€ 5.000	€ 0	€ 5.000,00
Organisatie 3	Open Access	€ 50.000									€ 25.000		€ 25.000		€ 50.000	€ 0	€ 50.000,00
Organisatie 4															€ 0	€ 0	€ 0,00
Ondernemingen																	
Onderneming 1															€ 0	€ 0	€ 0,00
Onderneming 2															€ 0	€ 0	€ 0,00
Onderneming 3															€ 0	€ 0	€ 0,00
Onderneming 4															€ 0	€ 0	€ 0,00
Overige (private) partners																	
Naam 1															€ 0	€ 0	€ 0,00
Naam 2															€ 0	€ 0	€ 0,00
Naam 3															€ 0	€ 0	€ 0,00
Naam 4															€ 0	€ 0	€ 0,00
Totaal			€ 105.000	€ 0	€ 0	€ 12.000	€ 0	€ 12.000	€ 0	€ 11.000	€ 0	€ 35.000	€ 0	€ 35.000	€ 0	Totaal kosten derden	€ 105.000

6.

Totale kosten	€ 4.327.160	€ 86.893	€ 15.300	€ 766.568	€ 61.200	€ 898.333	€ 61.200	€ 874.125	€ 61.200	€ 877.750	€ 61.200	€ 557.500	€ 45.900				€ 4.327.160
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FINANCIERING - Bijdragen per partner per jaar

In Euro		2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind		
Onderzoekorganisaties																	
1	Erasmus MC	€ 100.000		€ 400.000		€ 300.000		€ 400.000		€ 400.000		€ 400.000		€ 2.000.000	€ 0	€ 2.000.000	€ 0
2	EURP Rotterdam													€ 0	€ 0	€ 0	€ 0
3	TU Delft													€ 0	€ 0	€ 0	€ 0
...	Organisatie 4													€ 0	€ 0	€ 0	€ 0
Totale bijdrage onderzoekorganisaties		€ 100.000	€ 0	€ 400.000	€ 0	€ 300.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 400.000	€ 0	€ 2.000.000	€ 0	€ 2.000.000	€ 0
Ondernemingen																	
4	Onderneming 1													€ 0	€ 0	€ 0	€ 0
5	Onderneming 2													€ 0	€ 0	€ 0	€ 0
6	Onderneming 3													€ 0	€ 0	€ 0	€ 0
...	Onderneming 4													€ 0	€ 0	€ 0	€ 0
Totale bijdrage ondernemingen		€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0
Overige (private) partners																	
7	RSVA Erasmus MC	€ 7.935		€ 13.225										€ 21.160	€ 0	€ 21.160	€ 0
8	Rijndam		€ 15.300,00		€ 61.200,00		€ 61.200,00		€ 61.200,00		€ 61.200,00		€ 45.900,00	€ 0	€ 306.000	€ 306.000	€ 306.000
9														€ 0	€ 0	€ 0	€ 0
10														€ 0	€ 0	€ 0	€ 0
11														€ 0	€ 0	€ 0	€ 0
...	Naam 4													€ 0	€ 0	€ 0	€ 0
Totale overige bijdrage		€ 7.935	€ 15.300	€ 13.225	€ 61.200	€ 0	€ 61.200	€ 0	€ 61.200	€ 0	€ 61.200	€ 0	€ 45.900	€ 21.160	€ 306.000	€ 327.160	€ 327.160
		In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind	In cash	In kind			Totaal	
Totale Financiering per jaar		€ 107.935	€ 15.300	€ 413.225	€ 61.200	€ 300.000	€ 61.200	€ 404.000	€ 61.200	€ 407.000	€ 61.200	€ 407.500	€ 45.900	€ 2.327.160		€ 2.327.160	
Aan te vragen flazchip subsidie		€ 21.042	€ 0	€ 363.333	€ 0	€ 598.333	€ 0	€ 474.125	€ 0	€ 477.750	€ 0	€ 107.500	€ 0	€ 2.000.000			Funding gap
Totale budget		€ 86.893	€ 15.300	€ 766.568	€ 61.200	€ 898.333	€ 61.200	€ 874.125	€ 61.200	€ 877.750	€ 61.200	€ 557.500	€ 45.900	€ 4.327.160			



Fondsenwerving voor
het Erasmus MC-Sophia Kinderziekenhuis



Convergence for Health & Technology
Attn. Prof. dr. ir. R. Goossens

Subject: Letter of commitment

Rotterdam, February 28th 2022

Dear Mr Goossens,

The mission of the foundation *Vrienden van Sophia* is to support important projects in the field of research, innovation, welfare, and research in order to realize the best care for sick children. Therefore, we wholeheartedly declare our commitment to the realization of Child Brain Lab through a contribution of 400.000 euro per year for 2 years from already raised funds. We are pleased with the opportunity to bring in the necessary matching for the first two years of the Child Brain Lab flagship application.

Yours sincerely,


Laura van Roon
Stichting Vrienden van Sophia

Convergence for Health & Technology
Attn. Prof. dr. ir. R. Goossens

11 March 2022

LETTER OF COMMITMENT
for the Child Brain Lab Flagship

Dear Prof. Dr. Ir. R Goossens,

I, Evelyn Lindeman in my capacity of CEO Rijndam Revalidatie hereby confirm that Rijndam intends to contribute to the Child Brain Lab Flagship, as applied for by the main applicant, dr. Marie-Lise van Veelen, Head of the Child Brain Lab of Erasmus MC – Sophia Children's Hospital.

Rijndam is already actively collaborating with the Department of Rehabilitation Medicine, TU Delft, the EUR and the Child Brain Lab of Erasmus MC on the development of children with neurodisability on meaningful outcomes, on new techniques to track motor development and optimize mobility, and on meaningful and personalized care with patient participation.

We support these initiatives within the Child Brain Lab Flagship with in kind contributions of our clinical researchers in pediatric rehabilitation, representing a total amount of 306.400 Euro over a 5 year period. This in kind contribution is part of the so called 'vrije beleidsruimte' for their research tasks.

Yours sincerely,



Evelyn Lindeman
CEO Rijndam Rehabilitation



Erasmus MC
Attention to Dean, Prof. Dr. Stefan Sleijfer
Dr Molewaterplein 40
3015 GD ROTTERDAM

12 March 2022

LETTER OF COMMITMENT
for the
CHILD BRAIN LAB FLAGSHIP

Dear Dean Prof.Dr. Stefan Sleijfer,

I, Gerard Ribbers in my capacity of Head of the Department of Rehabilitation Medicine hereby confirm that the Department of Rehabilitation Medicine intends to contribute to the Child Brain Lab Flagship, as applied for by the main applicant, dr. Marie-Lise van Veelen, Head of the Child Brain Lab of Erasmus MC – Sophia Children's Hospital.

The Department of Rehabilitation Medicine intends to provide matching funding of a 0.5 fte post-doc (period: October 2022 – May 2023) to this Flagship. This post-doc is already actively collaborating with the TU Delft on a project to quantify trunk control in infants, and is externally funded by the Rijndam Rehabilitation Research Fund. Her dedicated research time (7 months period) represents a monetary value of € 21.160 and further detailed in the project proposal and budget form.

Yours sincerely,

Name: Prof. Dr. GM Ribbers

Position: Head of Department of Rehabilitation Medicine, Erasmus MC

Date: 12-3-2022



Convergence of Health and Technology

Direct dial (010) 70 35993

Fax number

Internal postal address Na 2118

E-mail a.onstenk@erasmusmc.nl

Our reference ML/2022

Date March 09, 2022

Concerning Letter of support from department head of main lead

Postal address

P.O. box 2040
3000 CA Rotterdam, NL

Visiting address

Dr. Molewaterplein 40
3015 GD Rotterdam, NL

Contact & route

www.erasmusmc.nl

Dear Madam, Sir,

Head of Department

Prof. dr. C.M.F. Dirven

Dr van Veelen has held the position of lead of the Sophia Pediatric Brain Center, since its start in 2019. The Child Brain Lab is the most important project within the Pediatric Brain Center. The realization of this initiative is progressing well and inspiring for all participating members.

I fully support Dr van Veelen as main lead for this flagship Child Brain Lab consortium. The granted in-kind support amounts to 0,1 fte.

Kind regards,

Prof. dr C.M.F. Dirven
Head of Department Neurosurgery

Neurosurgeons

Dr R.K. Balvers
Dr E.M. Bos
Drs. A.H.G. Dallenga
Dr. R. Dammers
Drs. E.J. Delwel
Drs. I.K. Haitsma
Dr. B.S. Harhangi
Drs. H. Hulscher
Drs. T.H.R. de Jong
Drs. H.P. van Putten
Drs. J.W. Schouten
Drs. J.K.H. Spoor
Dr. M.L.C. van Veelen
Dr. A.J.P.E. Vincent

Senior Researchers

Prof. dr. S. Leenstra
Dr. M.L. Lamfers
Dr. J. de Vrij